RAGHAVENDRA DINESH

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SUMMARY

Experienced Data Scientist and Deep Learning Engineer with a strong background in AI and machine learning. Successfully engineered applications using FastAPI and LLMs for contract data analysis and developed a custom BLIP-2 model for plant health diagnostics. Proven ability to enhance diagnostic precision and expedite proposal submissions. Eager to leverage expertise in generative AI to drive innovation and achieve strategic goals.

PROFESSIONAL EXPERIENCE

Indra Solutions LLC | *Staff Data Scientist* **May 2025 - Present**

- Led a cross-functional team to develop custom MCP (Model Context Protocol) agents, improving real-time data processing capabilities following Agile Practices on Azure DevOps
- Spearheaded the design and implementation of an organization-wide Intranet platform, enhancing collaboration and accessibility for all employees
- Mentored junior engineers in LLM app development, Supabase optimization, and AI pipeline deployment best practices, leading to improved project completion rates and technical skills development
- Re-engineered GitHub CI/CD workflows, reducing Azure Container costs through purging and image optimization, resulting in significant cost savings
- Deployed cost-efficient LLM ChatBots using Supabase edge functions and RPC calls, optimizing inference pathways and concurrency limits, enhancing user interaction speed and experience
- Engineered a Retrieval-Augmented Generation (RAG) pipeline using Supabase, LangChain and RPC functions, cutting query latency from 3 minutes to 12 seconds through optimized data retrieval/caching.
- Integrated OpenAI APIs for LLM inference and fine-tuned model selection, reducing API latency from 2–3 minutes to 30 seconds via prompt optimization and efficient token usage, achieving 70% cost reduction while maintaining model accuracy and responsiveness.

Arizona State University | *Deep Learning Engineer* **Jun 2024 - May 2025**

- Developed a custom BLIP-2 that integrates large vision and language models for plant health diagnostics, achieving 98% accuracy with a Lite ResNet50 model and 96% accuracy with a fine-tuned SWIN Transformer over 40 epochs
- Increased diagnostic precision by creating a system that uses QFormer to connect a frozen SWIN encoder with Llama 3.2 3B, achieving a maximum RougeL F1 score of 0.86 for generating localized cures
- Improved treatment insights by first validating a proof-of-concept using Gemini API (RougeL F1 score of 0.2) and later optimizing performance with PEFT on Vicuna 7B (score of 0.6) before transitioning to Llama 3.2 3B.

America Reads, Mary Lou Fulton Teachers College | *Tutor* **Oct 2022 - May 2024**

- Developed a Gemini-driven platform that automatically generated daily activity plans and personalized worksheets, improving student engagement and learning outcomes
- Enhanced adaptive assessment by designing a module that customizes questions based on student progress, leading to more accurate evaluations and tailored learning experiences

Bosch Global Software Technologies Pvt. Ltd. | *Automation Engineer* **Sep 2019 - Aug 2022**

- Delivered an RF emission analysis portal with interactive visualizations using Plotly. Applied Agile practices to refine an Autoencoder anomaly detection system, reducing analysis time from 30-50 hours to 6-8 hours and improving efficiency
- Advanced lab asset management by designing a smart inventory tracking system on Raspberry Pi 3B that integrated YOLOv4-tiny for asset detection and Tesseract OCR for auto labeling, to provide real-time tracking.
- Accelerated report generation, reducing manual processing time by 90% by building a web application to automate data extraction and document rendering.

ConnectM Technology Solutions Pvt. Ltd. | *Artificial Intelligence Intern* **Jan 2019 - Mar 2019**

- Developed a Driver Assistance System with Safety Constraints using Facial Recognition and Voice Assistance, enhancing driver safety by alerting them to potential hazards
- Built voice and face recognition systems using CMU Pocket Sphinx, Haar Cascade, and AWS Face Rekognition, improving system accuracy and user interaction
- Trained drowsiness detection using Facial Landmark detection and integrated all modules on Raspberry Pi 3B, resulting in a comprehensive system that alerts drivers to prevent accidents

EDUCATION

Arizona State University	*Masters, Artificial Intelligence (Robotics and Autonomous Systems)* (GPA: 3.97)	**Aug 2022 - May 2024**
PES Institute of Technology	*Bachelor of Engineering, Electronics and Communication Engineering* (GPA: 3.2)	**Aug 2015 - Jun 2019**

TECHNICAL SKILLS

- **Programming & Scripting:** Python, C++, React/JS, HTML/CSS, R, SQL
- **Interests:** Computer Vision, Deep Learning, Natural Language Processing, Generative AI
- **Tools:** Pandas, NumPy, Tensorflow, PyTorch, OpenCV2, Sklearn, Onnx, TFLite, GIT, Docker, FastAPI, Supabase, Azure, Pinecone, GitHub Actions, OpenAI API, Gemini API, LangChain

PROJECTS

Negative CLIPort: Robotic Arm enhanced with Negative Goals **2023**
Perception in Robots

- Improved CLIPort to handle a wider range of requirements by making it more robust to negative language goals.
- Training data made with actions the arm avoids, and loss function modified to penalize conflicting actions.

Medical Image Analysis for Lung Disease Detection **2023**
Medical Image Processing

- Leveraged Swin Transformer, Faster R-CNN, UPerNet, and U-Net for classification, localization, and segmentation of Chest X Rays from VinBigData CXR, SIIM-ACR Pneumothorax, MedMNISTv2 and MINIJSRT
- Developed a multi-task model to accurately analyze X-rays, achieving high score in metrics like AUC, IoU, FROC.